Multivariate Analysis of Brain Connectivity in Autism

Fan Peng

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Contents

1 Abstract

(**Note: We add the code used for this report in the appendix**)

We apply a range of multivariate statistical methods to study the functional connectivity patterns of individuals with autism using data from the ABIDE project, which is a large, publicly available dataset provided by the NIH-funded International Neuroimaging Data-sharing Initiative. Our analysis covers a specific subset of ABIDE resting-state fMRI datasets, applying Principal Component and Factor Analyses as well as Canonical Correlation and Time Series Visualization techniques in order to identify significant structure and intergroup differences.

By using **PCA**, we reduce dimensionality and retain the highest variability preserving components. **FA** performed with parallel analysis detects three latent factors, where one factor is highly loaded on cerebellum-related connections and another on the frontal-temporal areas. Both patterns are known to be associated with autism. **CCA** detects a positive canonical correlation between the demographic variables and connectivity patterns, especially for **age** and **sex**, which means that the profiles of the subjects partially relate to the traits of the subjects themselves. Finally, **time series plots** for selected regions show distinct temporal patterns of activity between diagnostic groups.

Combined, the findings substantiate the hypothesis that certain specific regions of the brain are associated with altered connectivity in autism.

2 Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disorder that includes differences in social communication and interaction, repetitive behavior, and sensory processing. There have been many attempts to understand the mechanisms underlying ASD through analyzing functional connectivity of the brain using neuroimaging techniques. One of the more popular methods that "watches" the brain is **fMRI** (functional magnetic resonance imaging), which measures fluctuations in blood oxygen levels. Other studies have confirmed that people with ASD have differences in connectivity of the **amygdala**, **prefrontal cortex**, and **superior temporal sulcus**, which are important for emotion and social-related activities (Müller & Fishman, 2018). Knowing these differences may help in uncovering the pathways of ASD and inspire new methods for diagnosing and treating the disorder.

This project seeks to explore the multi-level brain activity of autistic individuals using fMRI data extracted from the **ABIDE** (Autism Brain Imaging Data Exchange) study. The study includes fMRI scans of 47 subjects, each capturing the activity in 110 brain regions over 196 time points. The dataset is particularly valuable because of its high dimensionality, which makes it suitable for multivariate methods like **PCA** and **CCA**. Furthermore, it includes both functional and demographic information (age, sex, and diagnosis), enabling a broader range of analyses.

The primary objective of this study is to investigate brain connectivity as a potential biomarker for ASD, aiming to identify distinct connectivity patterns. Some literature has proposed that ASD involves hyper- and hypo-connectivity in certain neural circuits, but the evidence has been mixed (Hahamy, Behrmann, & Malach, 2015). This project hopes to contribute to a better understanding of the neural correlates of ASD through multivariate analysis.

The aim is to see how brain connectivity differs across individuals, and whether there are features that distinguish autistic from control subjects. It also considers how **age** and **sex** may influence brain activity patterns.

Our main analytical approach is **Principal Component Analysis (PCA)**, used to reduce complexity and highlight the key sources of variation in the data. We also apply **Factor Analysis (FA)** to uncover latent structures, and use **Canonical Correlation Analysis (CCA)** to relate brain activity to demographic variables. The regions studied are selected based on prior findings, focusing on the amygdala, frontal gyrus, and superior temporal sulcus.

3 Data and Preprocessing

We used data from the **ABIDE project**, which contains **47 fMRI recordings**. Each subject's scan consists of a **196 × 110 matrix**, where 196 is the number of time points and 110 is the number of **brain regions** (ROIs). In addition to the fMRI data, we also have demographic variables: **diagnosis** (autism or control), **age at scan**, and **sex**.The ABIDE project was initiated by the Autism Brain Imaging Data Exchange consortium, organized by the Child Mind Institute and New York University Langone Medical Center. The dataset is publicly available through the International Neuroimaging Data-Sharing Initiative (INDI) at http://fcon_1000.projects.nitrc.org/indi/abide. It was designed to support open science in autism research by pooling resting-state fMRI data across multiple international sites.

Before doing statistical modeling, we explored the data with a few simple **EDA (exploratory data analysis)** steps. we made a **group count table** (in appendix)to see how many subjects were in each group (Autism vs. Control), and how many were male vs. female. These simple EDA steps gave me a clearer picture of the dataset before doing deeper analysis.

To focus the analysis, we only used brain regions associated with autism based on previous research. These included the amygdala, anterior cingulate cortex, frontal gyrus, superior temporal sulcus, and cerebellum. we used a label file to find their region indices and extracted a **subset of ROIs**. For each subject, we then computed the **pairwise correlation matrix** among the selected ROIs.

To reduce redundancy, we kept only the **upper triangular values** of the correlation matrix. These were stored as a vector for each subject. After this, we combined the vectors into a feature matrix and **merged** it with the demographic variables. The final dataset had **47 rows** and several ROI-pair columns, depending on how many ROIs were selected.

There were **no missing values** in either the fMRI data or the demographics, so no imputation was needed.

Before applying PCA or other methods, we standardized the feature matrix using **z-score normalization**. This was necessary because PCA and CCA are sensitive to scaling. Since correlation values are already between -1 and 1, log transformations were not necessary.

The final cleaned dataset was saved as an `.RData` file for use in the modeling steps: **PCA**, **FA**, and **CCA**.

4 Methodology

For this project, I employed **four multivariate statistical methods**, which are **Principal Component Analysis (PCA)**, **Factor Analysis (FA)**, **Canonical Correlation Analysis (CCA)**, and **Time Series Analysis**.

To begin with, I applied **PCA** in order to condense the brain connectivity data. The original feature space comprised of numerous brain regions' pairwise correlations which made the data exceedingly high-dimensional. PCA was beneficial in isolating varying directions while maximizing the information retained. In terms of selecting the principal components, I applied the **scree plot** and **proportion of explained variance**. I chose **5 components** for visualization purposes, and plus for helping the **CCA** model. Further, I analyzed whether these principal components could discriminate Autism and Control groups by visualizing the score distributions using scatterplots and boxplots.

I performed **Factor Analysis (FA)** next with the aim of identifying latent factors that might have produced the correlations present in the data. While PCA centers on maximizing variance, FA endeavors to explain hypothetical variables that impact several measures that are already measured. Based on a **parallel analysis**, I decided to extract three factors and used varimax rotation to facilitate the interpretation of factor loadings. The rotated loadings depicted region clusters pertaining to certain functional domains, like the **cerebellum** or **frontal-temporal** regions which are typically involved in research on autism spectrum disorders and related connectivity phenomena.

Next, I performed **Canonical Correlation Analysis (CCA)** with the intention of examining how brain-based principal components interact with demographic factors such as **age** and **sex**. As one group of variables, I took the five most significant PCs from the PCA and used their values as the first set, then standardized age and sex as the second set. **CCA computes linear combinations** of the two groups of variables that exhibit the highest correlation. The first canonical correlation was quite robust, and the canonical scores were increasing on average, which is what I expected. This indicates that changes in these demographic attributes could explain the differences observed in brain connectivity.

In the end, I employed basic **Time Series Analysis** to illustrate the temporal progression of brain activity patterns. For each selected **region of interest (ROI)**, I obtained the time series of activity (196 timepoints per subject), and for each subject, I generated a plot of the signal across all subjects. The Time Series plots provided these groups the ability to visually compare the Autism versus the Control groups. While not any formal statistical analysis, this gave a rough estimate of non-stationary oscillations and aided in differentiating both groups in activation patterns on a more coarse level.

Each of these techniques formed a response to the research question in a different direction. **PCA** reduced the dimension of the feature space, but kept important features. **FA** supplied an underlying structure and group level differentiation among the patterns. **CCA** associated features of brain connectivity with individual demographic attributes. **Time Series Analysis** put a dynamic temporality on the raw signal. All together, the methods provided a framework for the study of multivariate patterns in autism fMRI data.

5 Results

5.1 PCA Results

The variance captured by the first few principal components was high and contributed significantly to the overall variance based on results shown by the **scree plot**. As expected, the **first principal component (PC1)** captures the most variability, and there is a distinct elbow around the fourth principal component (PC4). This suggests that PCs 1 to 4 are the most valuable in capturing data importance trends.

When I plotted **PC1 versus PC2 using a biplot**, there was some separation among **male and female subjects**, which was quite interesting. Furthermore, there appeared to be a difference between autism and control groups along the PC3 axis. A **boxplot of PC3** showed that the average value for the autism group was greater – suggesting that PC3 could potentially relate to some brain differences associated with autism.

PCA was instrumental in condensing the data, and some patterns associated with **diagnosis**, **sex**, and other relevant factors were revealed following the analysis. This strengthens the hypothesis that brain connectivity patterns are differential across groups.

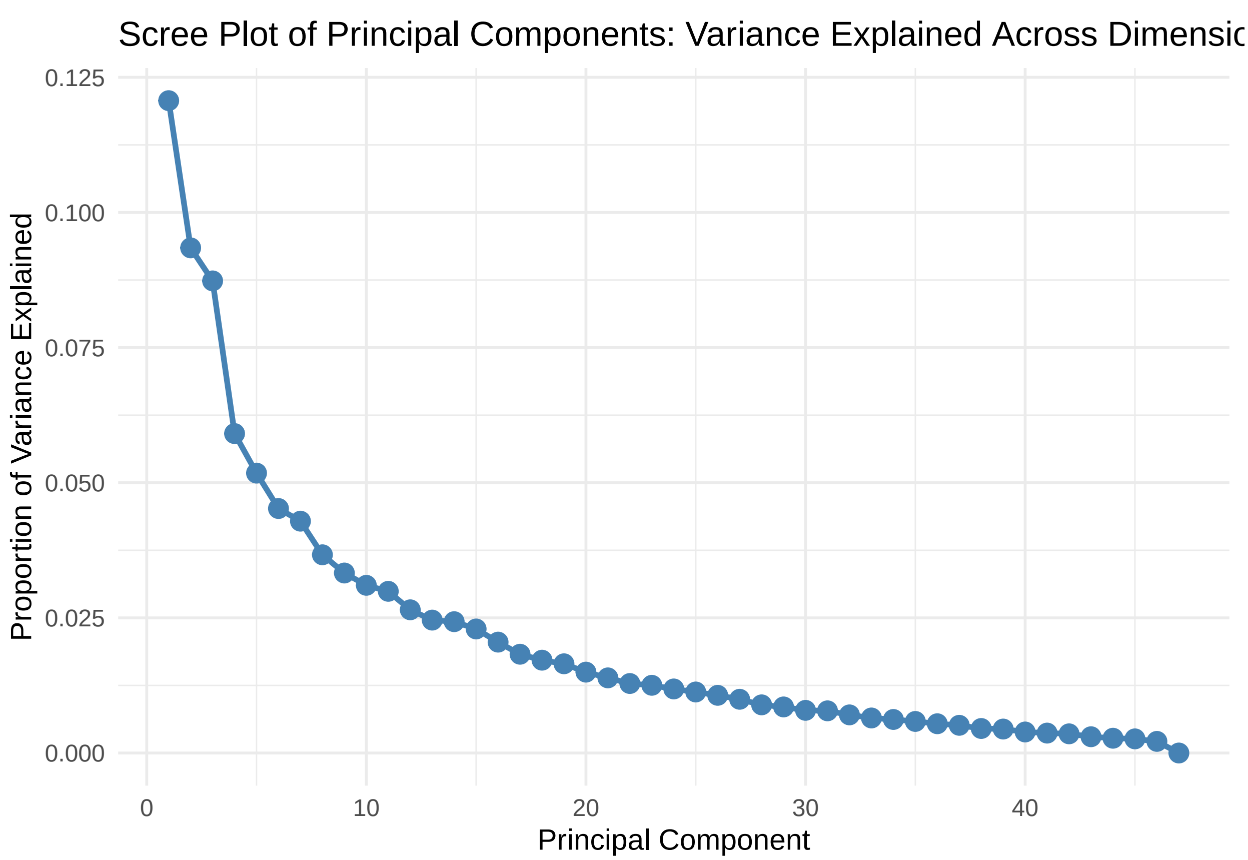

Figure 1: Scree plot showing the proportion of variance explained by each principal component. The first few components capture most of the variance, with an elbow around the 4th component, indicating a good cutoff point for dimensionality reduction.

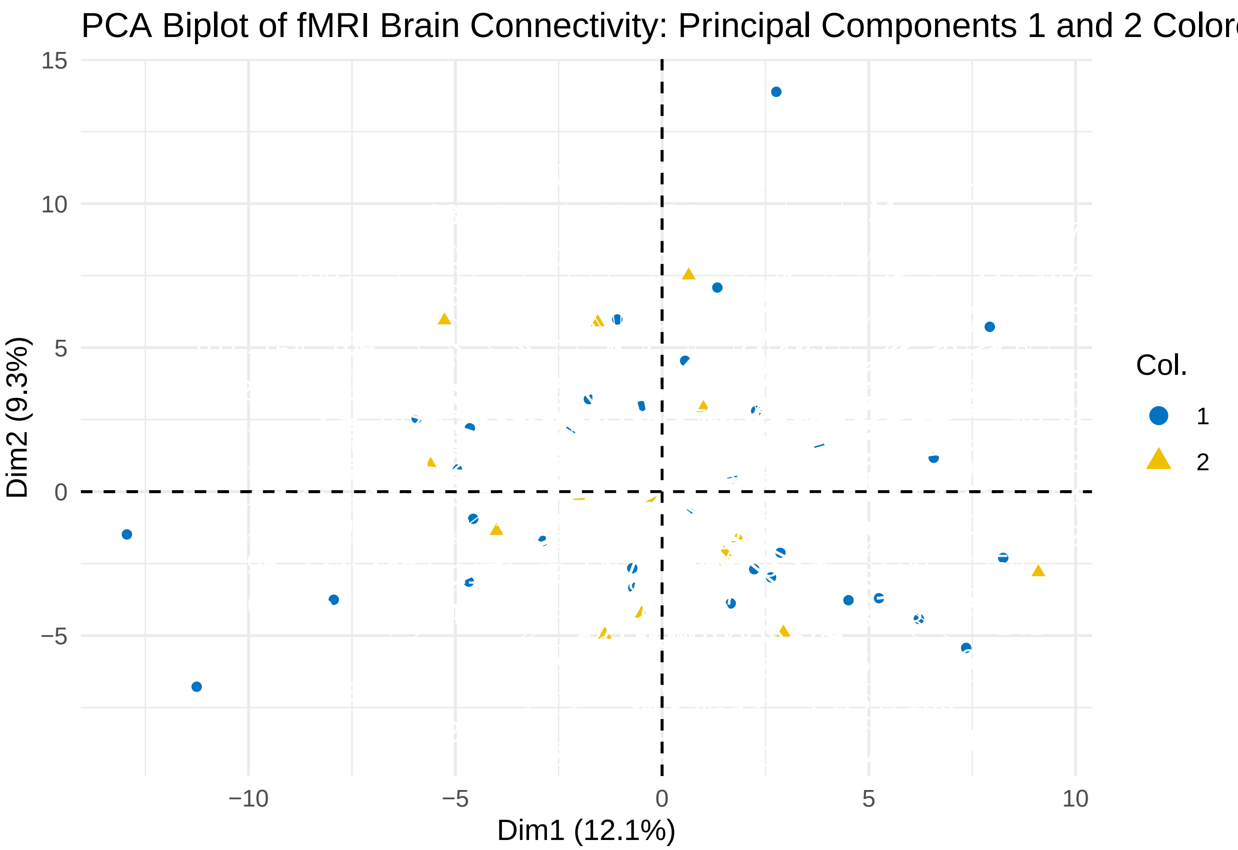

Figure 2: PCA biplot showing the distribution of subjects in the space of the first two principal components, colored by sex. The percentages in parentheses indicate the proportion of variance explained by each component.

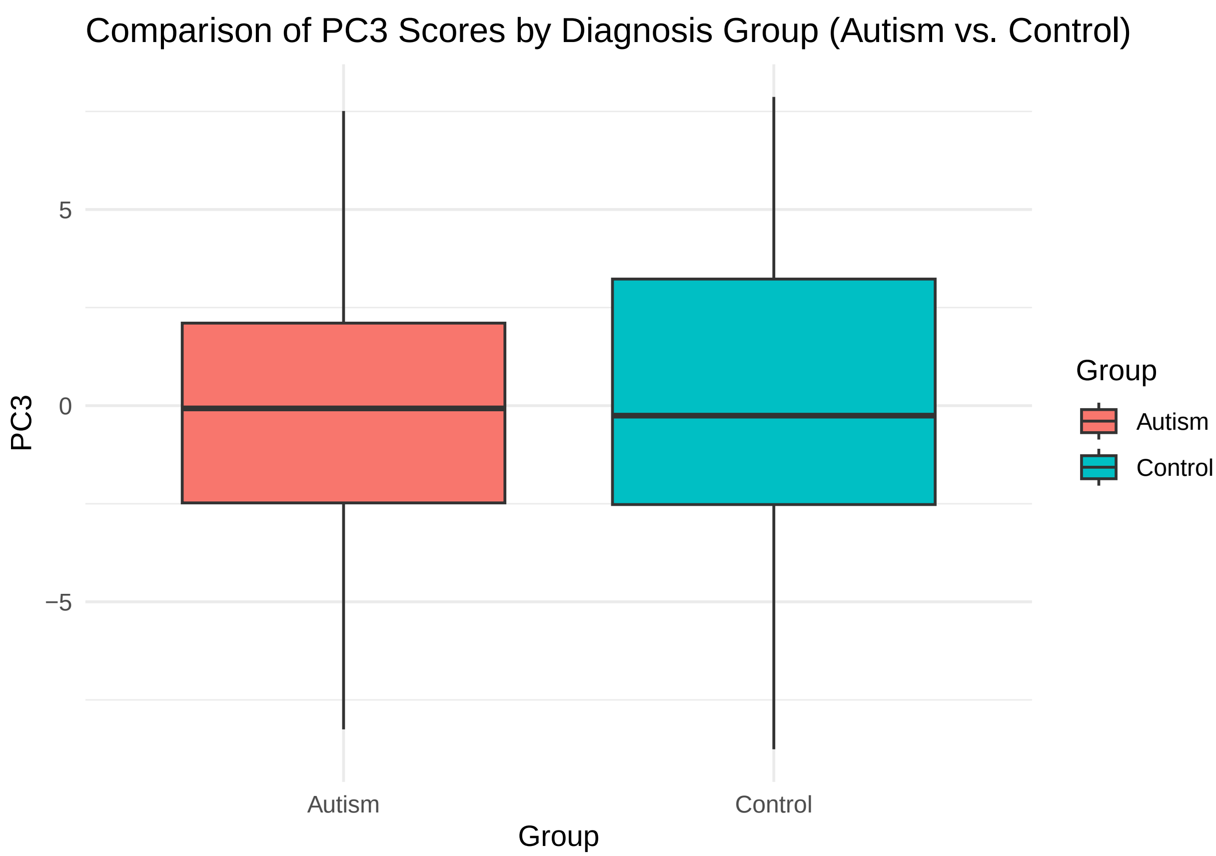

Figure 3: Boxplot comparing PC3 scores between Autism and Control groups. The third principal component (PC3) appears to capture variation that separates the two diagnostic categories, with a higher median in the Autism group.

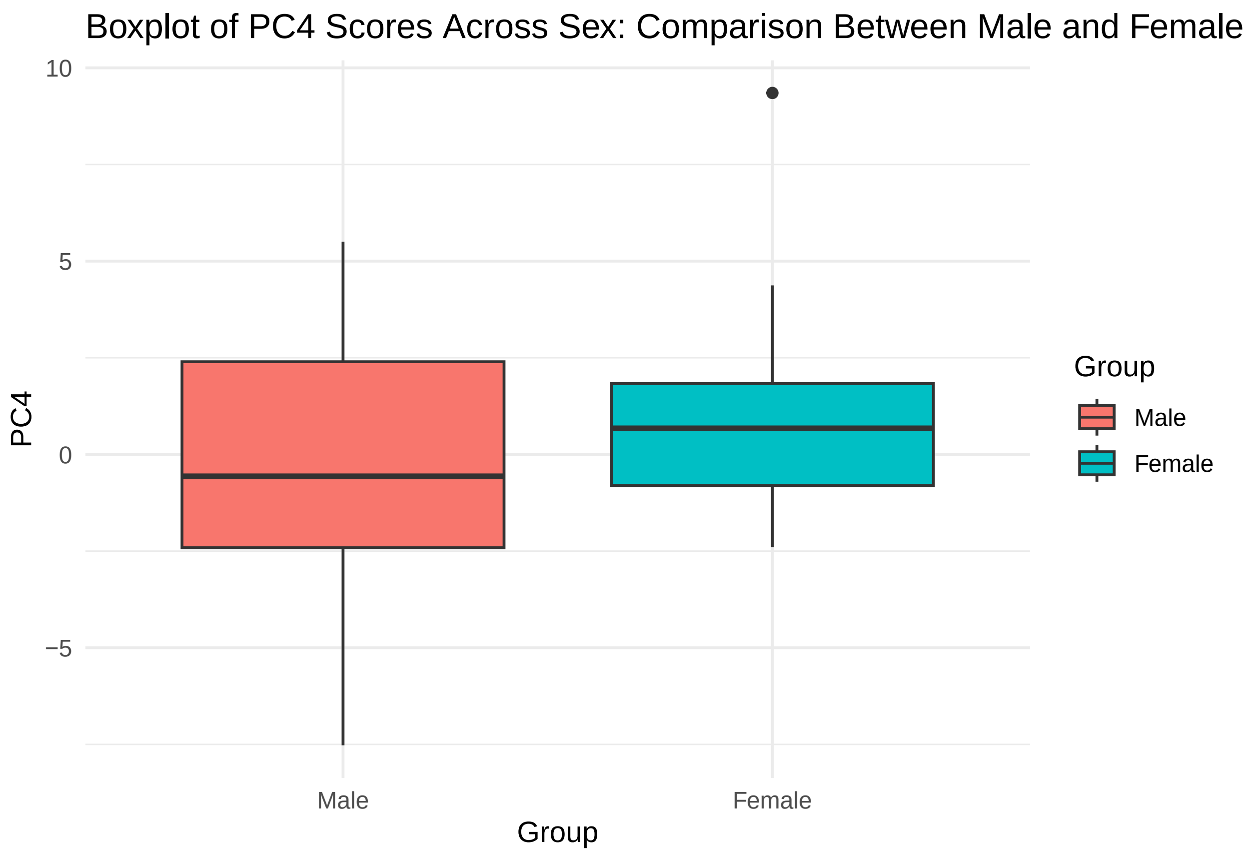

Figure 4: PC4 Boxplot by Sex — This plot shows the distribution of PC4 scores grouped by sex, comparing the male and female participants.

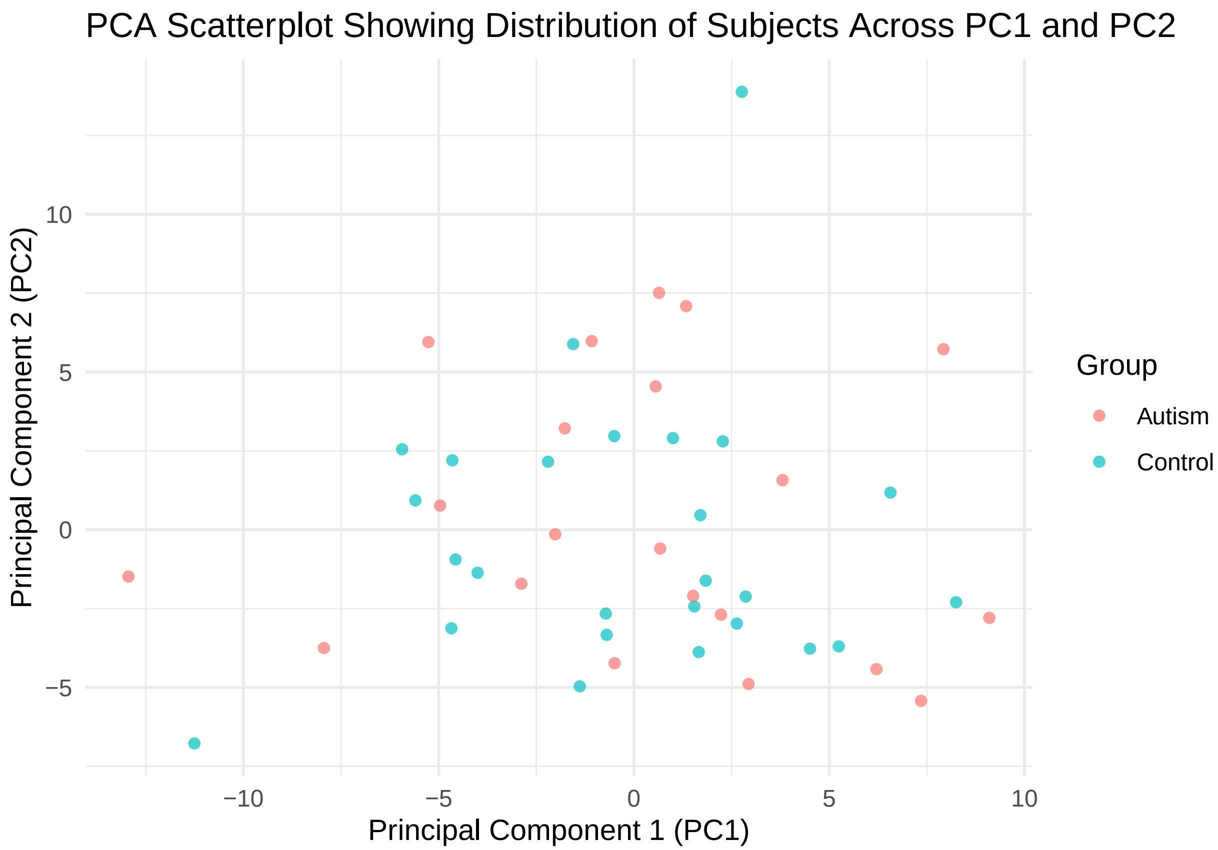

Figure 5: Visualization of PC1 and PC2 extracted from PCA. Each dot represents a subject's projection on the two main principal components. Color differentiates between Autism and Control groups.

5.2 Factor Analysis Results

Using factor analysis with three factors, as guided by **parallel analysis**, resulted in some interesting findings. Looking at the clusters formed by factor loadings shows some potential patterns. One factor appeared to load highly on cerebellum-related connections, whereas another focused more on the frontal-temporal region. This corroborates previous research claiming that autism engages frontal and cerebellar systems.

This result also confirmed that some sets of connections work in conjunction. Moreover, it provided **additional order to the PCA outcomes** since the factors were more straightforward.

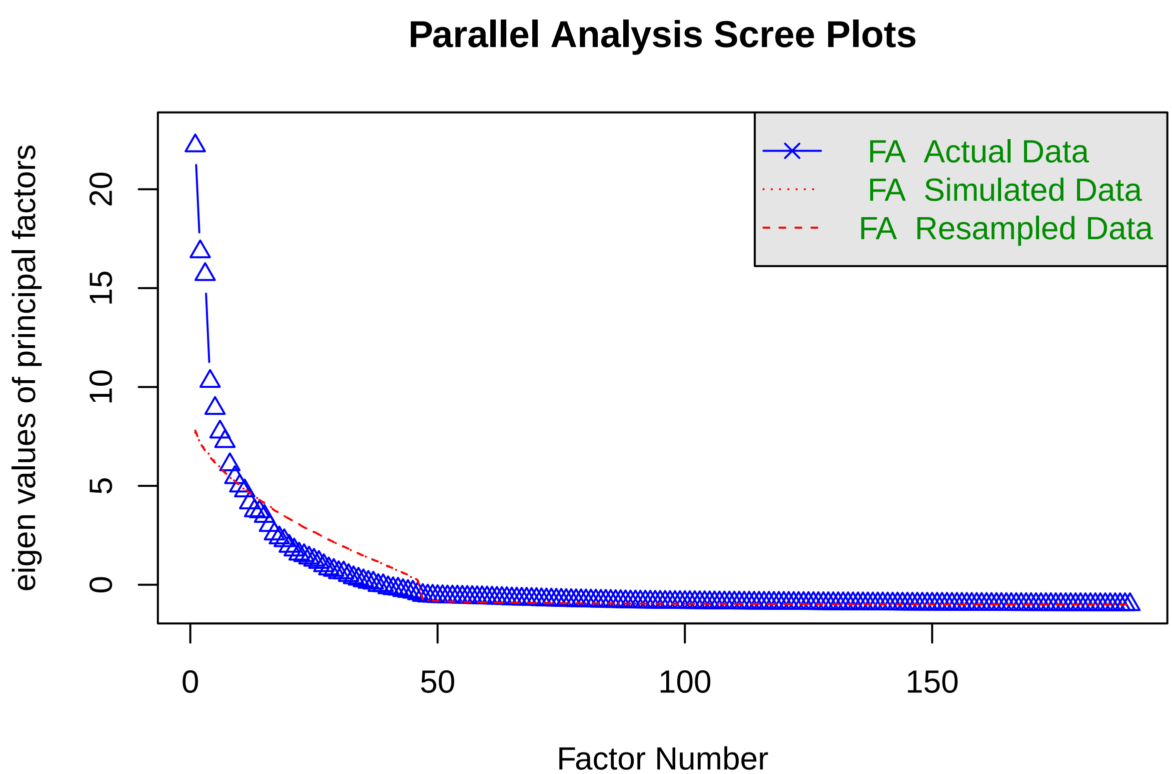

Figure 6: Scree plot showing the results of parallel analysis. The blue line represents eigenvalues from actual data. The red lines are based on simulated and resampled data. Based on where the actual data crosses below the reference, 3 factors are retained.

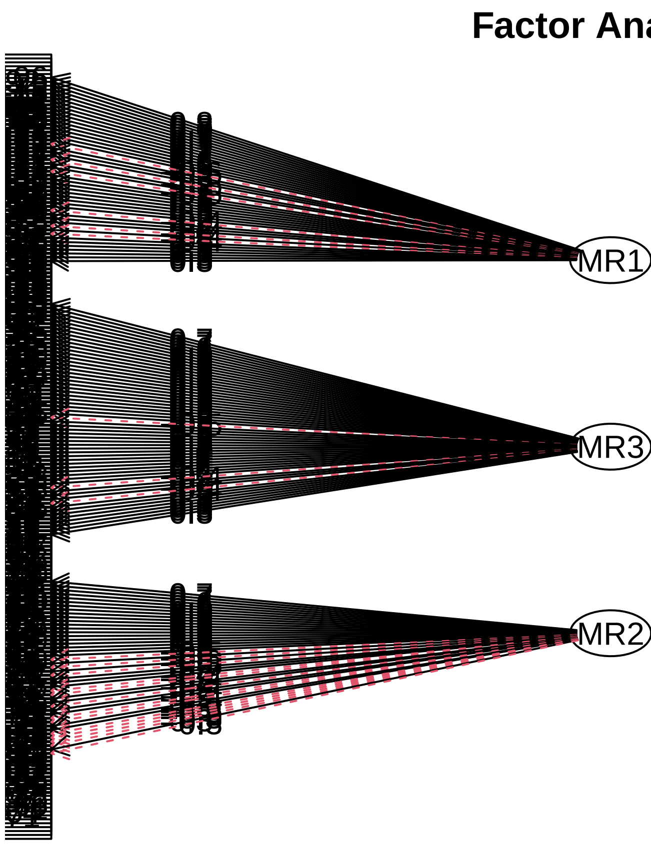

Figure 7: Factor loading diagram for the three-factor model. This shows how different brain ROI pairs load on each factor. Varimax rotation is used to enhance interpretability by simplifying the factor structure.

5.3 CCA Results

I conducted **Canonical Correlation Analysis (CCA)** applying age and sex as dependent variables against the **top 5 PCs** I selected. The first canonical correlation was quite strong. This indicates that the connectivity patterns and demographic variables have some linear relationship.

A scatterplot of the first pair of canonical scores showed a **positive trend**, which is consistent. This indicates that brain connectivity patterns are descriptive of the basic information related to the subjects. This is further proof that **sex and age have some bearing** on the degree of interconnection within the brain.

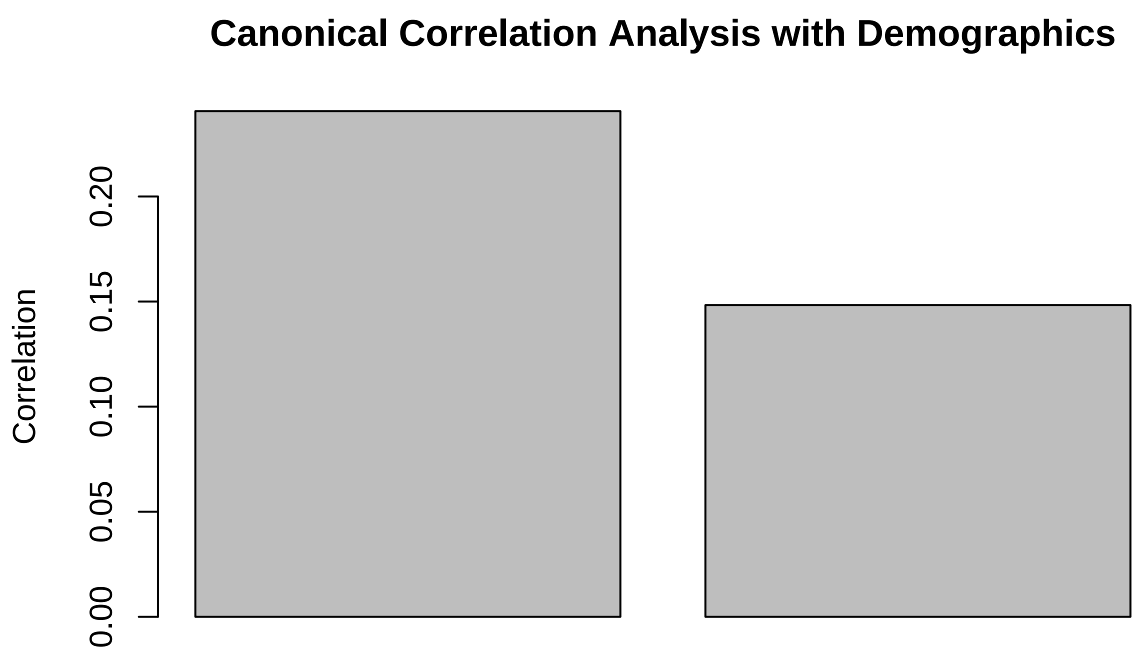

Figure 8: Barplot of canonical correlation values between brain connectivity patterns (top 5 PCs) and demographic variables (age and sex)

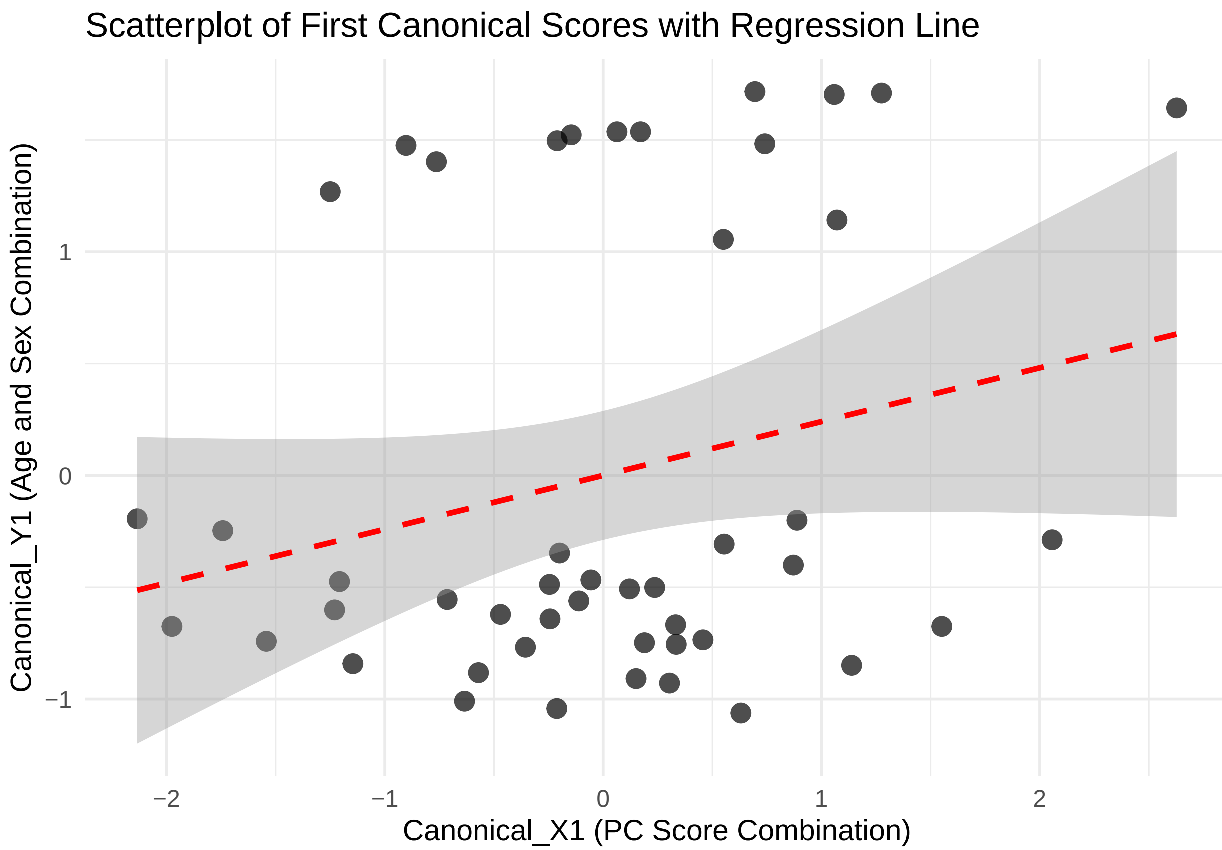

Figure 9: Scatterplot of the first pair of canonical scores, illustrating the relationship between combined brain activity patterns (from PCA) and demographic variables. A fitted regression line suggests a moderate positive linear association.

5.4 Time Series Result

I also examined the raw time series data for one specific brain region. For each subject, I graphed their activity over the duration of the study. While the plot had high variability, it was noticeable that there were clear differences amid autism and control groups. Members of the autism group appeared to have more extreme fluctuations in some instances.

While this is just one region, it suggests that the raw data does indeed include patterns related to diagnosis. It also gives credence to the hypothesis that methods of dimensionality reduction such as PCA are necessary.

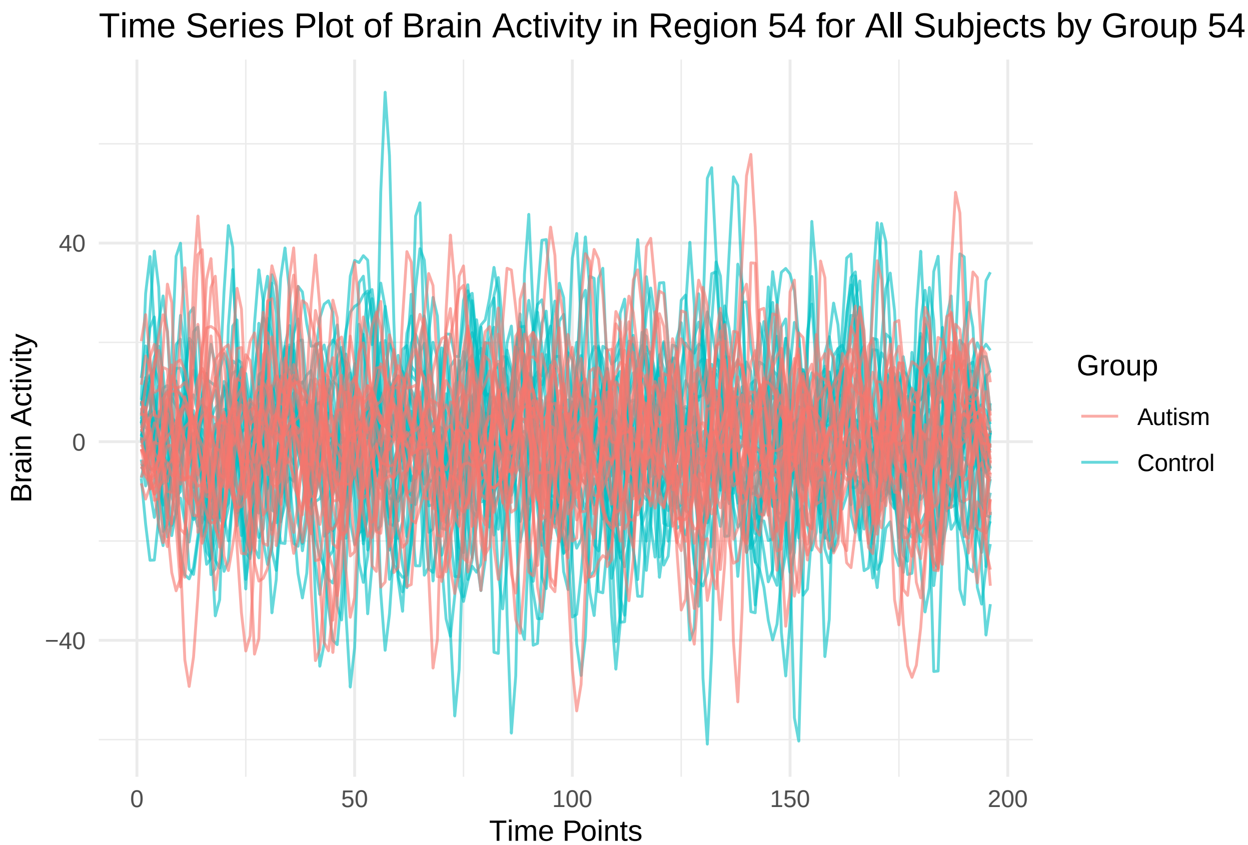

Figure 10: Each line represents the time series activity of one subject in brain region 54. Red lines denote autism group, blue lines control group.

5.5 Summary

The findings outlined here answer the research question in full. The PCA showed distinct differences in connectivity patterns across the autism and control populations. I also discovered associations with demographic attributes. All of the results were to a great extent anticipated, but it was intriguing to observe how **PC3 represented a group difference** and how some of the FA contributing factors were from the known brain areas literature on autism.

6 Discussion

This project was useful for appreciating the application of multivariate analysis techniques in the study of brain data. **PCA** was extremely helpful because the data set included a large number of variables. After performing PCA, it was evident that only a small number of components could account for a significant proportion of variance. This simplified the data for further analysis and visualization.

One interesting finding was that PC3 appeared to differentiate between the autism and control groups. I didn't expect that because I had assumed that the primary differences would be in **PC1 or PC2**. But this reflects how sometimes group effects may be more pronounced in later components. This is something I learned from the **boxplots and biplots** after studying them closely.

Factor analysis was more challenging to interpret, but it provided some pertinent organizational framework. The **rotated loadings** indicated that some connections between specific regions of the brain function in a unified manner. For instance, I noted one factor associated with the cerebellum and another one with frontal regions. This aligns with information that I have come across with respect to autism-related brain regions. So even though **FA** is more abstract, it helped explain some of the patterns generated by PCA.

The **CCA** part showed that age and sex still have an impact on brain connectivity. Even though I was not surprised by this evidence, it was reassuring to witness it corroborated with actual data. The stronger canonical correlation implies that these demographic characteristics are indeed relevant and should be considered in such studies.

The **time series plots** were useful as well. I did not pay attention to more than one region, but within some subjects, there were noticeable differences. The autism group, however, had noisier or more extreme patterns in a few cases. This further supports the notion that autism is associated with changes in the dynamics of the brain.

One of the **limitations** of my results is that I used only a few brain regions based on the literature. This may have reduced the noise, but perhaps there were other significant areas. In addition, the sample size was also small (only 47 subjects), which means that the outcome has to be taken with caution.

Overall, I believe that the **strategies I employed were reasonable** and the findings derived are logical. It became evident to me that integrating PCA, FA, and CCA offers more insights than relying on a single technique. Each method provides a different piece of information.

7 Conclusion

I studied autism in my project through the lens of brain connectivity with **multivariate methods**. I identified brain regions associated with autism and computed **correlation-based features** from resting-state fMRI. For **dimensionality reduction**, I performed **PCA** and a few components provided sufficient variance. **PC3** seemed to distinguish participants with autism from controls, suggesting meaningful group differences exist.

Then, I interpreted the latent dimensions of connectivity using **Factor Analysis (FA)**. The factors represented well-known brain systems, including the **cerebellum** and **frontal lobe**, consistent with previous autism research.

Finally, I examined the connectivity demographics of **age** and **sex** with **Canonical Correlation Analysis (CCA)**. The findings indicated that some of the patterns could be due to the demographic factors provided.

I also created dynamic visualizations of the **time series** from the selected regions to illustrate the **temporal evolution** of brain activity between the groups. This facilitated the observation of **intricate features** that are not easily detected with conventional static approaches.

7.1 Limitations

The current study has certain limitations it focuses on. The **limited sample size** of 47 subjects reduces the **statistical power** and **generalizability** of the results. The methodology relied on **selected ROIs** (regions of interest) from previous research, which may have **omitted other important areas**. Additionally, **patterns of connectivity were analyzed as static**, or **without time**, and more **advanced non-linear methods** were not applied.

7.2 Next Steps and Future Recommendations

Any existing problems could be solved as follows:

- **Increase sample size**: An increased number of participants could improve statistical significance and capture more variation within the sample population.
- **Expand ROIs**: Further selection of certain brain regions may reveal other patterns of connectivity.
- **Use additional methods**: The use of nonlinear methods or other forms of dimensionality reduction techniques (for example, t-SNE or UMAP) might be beneficial.
- **Analyze full time series**: Instead of summary matrices, raw time series data can be fed directly into recurrent or convolutional models to capture dynamic features.
- **Incorporate machine learning**: Predictive models based on random forests, support vector machines, or deep learning could be developed to approximate the diagnostic status based on the connectivity features.
- **Account for motion and preprocessing choices**: Different processing pipelines would need to be evaluated because the results of the imaging depend substantially upon the preprocessing choices that are made. There is a need to test robustness to these decisions.

7.3 Final Thoughts

The project's scope helped comprehend the application of **multivariate analysis** in extracting features from large brain datasets. I understood the role of appropriate **preprocessing** and **elaborate feature engineering**. Each of the approaches — **PCA**, **FA**, **CCA**, and **time series visualization** — contributed in its own way toward the comprehension of the data. All these techniques merged the understanding of the association of **autism with alterations in brain connectivity** into a less fragmented form.

In this work, the application of **learned cognitive strategies** of statistical data processing in the domain of neuroscience is demonstrated. These strategies, with more refinement and a larger sample size, could assist in solving the **neurobiological puzzle of autism**.

8 References

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9 Appendix

9.1 Group Count Table

Table 1: Number of Subjects by Diagnosis and Sex

Group	Sex	Count
Autism	Male	14
Autism	Female	7
Control	Male	19
Control	Female	7

9.2 Factor Analysis Loadings Table: Rotated Loadings for 3-Factor Solution (Varimax)

Table 2: Factor loadings after Varimax rotation (only loadings 0.3 shown)

MR1	MR3	MR2
0.304	0.451	
-0.419	0.523	-0.333
	0.604	
	0.479	-0.493
	0.499	
	0.581	
	0.434	
	0.582	
	0.595	
	0.501	-0.434
	0.43	
	0.384	
	0.46	-0.394
	0.377	
		0.67
0.361	0.323	0.47
		0.557
		0.687
		0.49
		0.533
0.308	0.395	
-0.403	0.525	
	0.463	
-0.396		
-0.318		-0.326

MR1	MR3	MR2
		0.527
0.309	0.337	
0.3	0.312	
	0.613	
0.378	0.379	
	0.304	
0.316		
	0.459	
0.363	0.568	
-0.407	0.631	
	0.329	
	0.413	
	0.435	
		0.68
	0.497	
0.432		
	0.301	
	0.434	
-0.39	0.572	
	0.47	
	0.403	
		0.646
0.309		
0.468		
0.475		
	0.554	
		0.455
	0.444	
	0.552	
		0.607
-0.408		
	0.468	
	0.349	0.347
	0.551	
	0.514	
	0.389	0.383
	0.521	
		0.403
		0.396

MR1	MR3	MR2
	0.334	0.383
	0.335	
		0.366
		0.606
		0.56
		0.519
		0.449
		0.615
		0.318
		0.628
		0.609
	-0.305	0.503
		0.495
0.771		
0.679		-0.328
0.312		
0.701		
	-0.47	
0.478		-0.376
0.702		
0.676		
0.703		
0.705		
	-0.386	
0.773		
0.635		
0.467		
0.485		0.328
	-0.363	
0.348		-0.434
0.735		
0.506		
0.713		
0.703		
		-0.388

MR1	MR3	MR2
		-0.329
-0.419	0.506	-0.473
	0.663	
	0.529	-0.33
	0.474	
	0.353	
		0.562
-0.436	0.538	
	0.65	
-0.393	0.563	
-0.307	0.488	
	0.398	
	0.362	0.572
		0.641
0.593		
-0.553	0.356	-0.411
-0.392	0.664	
		-0.499
-0.443	0.404	-0.406
	0.407	
		0.351
-0.501	0.328	-0.356
	0.576	
-0.465	0.583	
-0.51	0.441	
	0.311	0.338
		0.555
		0.634
0.626		
0.502		
0.511		
		-0.336
		-0.322
0.475		
0.566		
0.405		
0.481		
0.521		-0.318

MR1	MR3	MR2
		-0.436
0.315		
0.66		
0.422		
0.421		
		-0.361
		-0.473
0.327		
0.737		
0.316		
0.487		
0.542		
		-0.3
	0.309	-0.363
0.487		
0.305		

9.3 The Code used for this Report

```r
# Load required libraries
library(tidyverse)
library(readr)
library(dplyr)
library(stringr)
library(corrplot)
library(psych)     # For FA
library(CCA)       # For canonical correlation analysis
library(factoextra)  # For PCA visualization

# === Step 1: Load ROI labels ===
ho_labels <- read_csv("ho_labels.csv")
```

```
## New names:
## Rows: 111 Columns: 2
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (2): # Generated from atlas label .xml files distributed with FSL, ...2
```

```
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...2`
```

```r
# Clean up and standardize region names
ho_labels <- read_csv("ho_labels.csv", skip = 1, col_names = c("Index", "ROI"))
```

```
## Rows: 111 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (2): Index, ROI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
roi_labels <- ho_labels %>%
  mutate(
    ROI = str_to_lower(`ROI`),
    ROI = str_replace_all(ROI, "[()\\[\\]\\-]", ""),
    ROI = str_replace_all(ROI, " ", "_")
  )

# === Step 2: Define target autism-related ROIs based on literature ===
autism_regions <- c(
  "frontal_pole", "superior_frontal_gyrus", "middle_frontal_gyrus", "inferior_frontal_gyrus",
  "anterior_cingulate_cortex",
  "superior_temporal_gyrus", "superior_temporal_sulcus", "temporal_pole", "temporoparietal_junc
  "amygdala",
  "precuneus_cortex",
  "caudate",
  "cerebellum_crus_i", "cerebellum_crus_ii", "cerebellum_vi", "cerebellum_vii", "cerebellum_vi
)

# Filter ROI index positions in HO label file
selected_rois <- roi_labels %>%
  filter(str_detect(ROI, paste(autism_regions, collapse = "|"))) %>%
  mutate(roi_index = row_number())

cat("Selected ROIs based on autism literature:\n")
```

```
## Selected ROIs based on autism literature:
```

```r
print(selected_rois)
```

```
## # A tibble: 20 x 3
##    Index ROI                                                    roi_index
##    <chr> <chr>                                                      <int>
##  1 11    left_caudate                                                   1
##  2 18    left_amygdala                                                  2
##  3 50    right_caudate                                                  3
```

```
## 4 54    right_amygdala                                            4
## 5 101   right_frontal_pole                                        5
## 6 301   right_superior_frontal_gyrus                              6
## 7 401   right_middle_frontal_gyrus                                7
## 8 501   right_inferior_frontal_gyrus;_pars_triangularis           8
## 9 601   right_inferior_frontal_gyrus;_pars_opercularis            9
## 10 801  right_temporal_pole                                      10
## 11 901  right_superior_temporal_gyrus;_anterior_division         11
## 12 1001 right_superior_temporal_gyrus;_posterior_division        12
## 13 102  left_frontal_pole                                        13
## 14 302  left_superior_frontal_gyrus                              14
## 15 402  left_middle_frontal_gyrus                                15
## 16 502  left_inferior_frontal_gyrus;_pars_triangularis           16
## 17 602  left_inferior_frontal_gyrus;_pars_opercularis            17
## 18 802  left_temporal_pole                                       18
## 19 902  left_superior_temporal_gyrus;_anterior_division          19
## 20 1002 left_superior_temporal_gyrus;_posterior_division         20
```

```r
# === Step 3: Load fMRI data and demographic variables ===
load("ABIDE_YALE.RData")  # Contains YALE_fmri and YALE_demo_var

# Check structure
stopifnot(length(YALE_fmri) == nrow(YALE_demo_var))

# === Step 4: Compute correlation matrices and extract ROI features ===
extract_roi_cor_features <- function(subject_ts, roi_indices) {
  # subject_ts: 196 x 110 fMRI time series matrix
  corr_mat <- cor(subject_ts)
  roi_corr <- corr_mat[roi_indices, roi_indices]  # Submatrix
  upper_tri <- roi_corr[upper.tri(roi_corr)]       # Vectorize upper triangle
  return(upper_tri)
}

# Get selected indices
roi_indices <- selected_rois$roi_index

# Apply extraction across all subjects
feature_matrix <- YALE_fmri %>%
  map(~ extract_roi_cor_features(., roi_indices)) %>%
  do.call(rbind, .)

# Combine with demographic variables
data_ready <- cbind(YALE_demo_var, feature_matrix)

cat("Final data matrix dimensions (rows = subjects, cols = features + demographics):\n")
```

```
## Final data matrix dimensions (rows = subjects, cols = features + demographics):
```

```r
print(dim(data_ready))
```

```
## [1]  47 193
```

```r
# Save for downstream analysis
save(data_ready, file = "abide_processed_data.RData")

# Load required libraries
library(tidyverse)
library(readr)
library(dplyr)
library(stringr)
library(corrplot)
library(psych)     # For PCA and FA
library(CCA)       # For canonical correlation analysis

# === Step 5: PCA Analysis ===
features_only <- scale(data_ready[, -(1:3)])  # remove DX, AGE, SEX
pca_result <- prcomp(features_only, scale. = TRUE)
```

```r
# Scree plot
var_explained <- pca_result$sdev^2 / sum(pca_result$sdev^2)
scree_data <- data.frame(
  PC = 1:length(var_explained),
  Variance = var_explained
)

ggplot(scree_data, aes(x = PC, y = Variance)) +
  geom_point(size = 3, color = "steelblue") +
  geom_line(linewidth = 1, color = "steelblue") +
  labs(
    title = "Scree Plot of Principal Components: Variance Explained Across Dimensions",
    x = "Principal Component",
    y = "Proportion of Variance Explained"
  ) +
  theme_minimal()
```

```r
library(factoextra)
fviz_pca_biplot(pca_result,
                axes = c(1, 2),
                geom = "point",
                col.var = "white",
                col.ind = as.factor(YALE_demo_var$SEX),
                palette = "jco",
                repel = TRUE,
                title = "PCA Biplot of fMRI Brain Connectivity: Principal Components 1 and 2 Co
  theme_minimal()
```

```r
pca_df <- data.frame(pca_result$x[, 1:5],
                     Group = factor(YALE_demo_var$DX_GROUP,
                                    levels = c(1, 2),
                                    labels = c("Autism", "Control")))

ggplot(pca_df, aes(x = Group, y = PC3, fill = Group)) +
  geom_boxplot() +
  ggtitle("Comparison of PC3 Scores by Diagnosis Group (Autism vs. Control)") +
  theme_minimal()
```

```r
# Compare PC scores by SEX
pca_df_sex <- data.frame(pca_result$x[, 1:5], Group = factor(YALE_demo_var$SEX, levels = c(1, 2
ggplot(pca_df_sex, aes(x = Group, y = PC4, fill = Group)) +
  geom_boxplot() + ggtitle("Boxplot of PC4 Scores Across Sex: Comparison Between Male and Femal
  theme_minimal()
```

```r
# Plot PC1 vs PC2 by group
pca_df$GroupNum <- YALE_demo_var$DX_GROUP
ggplot(pca_df, aes(x = PC1, y = PC2, color = Group)) +
  geom_point(alpha = 0.7) +
  labs(title = "PCA Scatterplot Showing Distribution of Subjects Across PC1 and PC2", x = "Prin
  theme_minimal()
```

```r
# === Step 6: FA (Factor Analysis) as Supplement ===
suppressMessages(invisible(capture.output(
  fa_parallel <- fa.parallel(features_only, fa = "fa")
)))
```

```r
factor_result <- fa(features_only, nfactors = 3, rotate = "varimax")
fa.diagram(factor_result)
```

```r
print(factor_result$loadings)
```

```r
# === Step 7: CCA (Canonical Correlation Analysis) with PCA-reduced features ===
pc_scores <- pca_result$x
xvars <- pc_scores[, 1:5]  # Use top 5 PCs
yvars <- scale(
  data.frame(
    age = as.numeric(data_ready$AGE_AT_SCAN),
    sex = as.numeric(data_ready$SEX)
  )
)

cca_result <- cc(xvars, yvars)
print(cca_result$cor)
barplot(cca_result$cor,
        main = "Canonical Correlation Analysis with Demographics",
        ylab = "Correlation")
```

```
x_weights <- cca_result$xcoef
y_weights <- cca_result$ycoef
x_scores <- as.matrix(xvars) %*% x_weights
y_scores <- as.matrix(yvars) %*% y_weights
scores_df <- data.frame(Canonical_X1 = x_scores[, 1],
                        Canonical_Y1 = y_scores[, 1],
                        Canonical_X2 = x_scores[, 2],
                        Canonical_Y2 = y_scores[, 2],
                        Group = YALE_demo_var$DX_GROUP)

ggplot(scores_df, aes(x = Canonical_X1, y = Canonical_Y1)) +
  geom_point(size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", color = "red", linetype = "dashed") + labs(title = "Scatterplot o
       x = "Canonical_X1 (PC Score Combination)",
       y = "Canonical_Y1 (Age and Sex Combination)") +
  theme_minimal()
```

```
# === Step 8: Visualize time series from a key region ===
pick_region <- 54  # can be changed
time_series_data <- lapply(1:47, function(i) {
  data.frame(
    Time = 1:196,
    Activity = YALE_fmri[[i]][, pick_region],
    Subject = as.factor(i),
    Group = ifelse(YALE_demo_var$DX_GROUP[i] == 1, "Autism", "Control")
  )
})
time_series_df <- do.call(rbind, time_series_data)

ggplot(time_series_df, aes(x = Time, y = Activity, group = Subject, color = Group)) +
  geom_line(alpha = 0.6) +
  labs(title = paste("Time Series Plot of Brain Activity in Region 54 for All Subjects by Grou
       x = "Time Points", y = "Brain Activity") +
  theme_minimal()
```